

Study Area - Data

Dataset type	Recommended source/provider	Use in manuscript
Airborne magnetic / VRTP	Gawler Craton airborne survey data packages, SARIG	Magnetic predictors; Werner/PSO inversion input
Digital elevation / DEM	Gawler Craton airborne survey data packages or DEM source used	Slope, curvature, roughness
Lithology/geology	100K Geology – surface geology, SARIG	Lithological predictor
Faults/lineaments	100K Geology – linear structures, SARIG	Structural predictors; distance/density layers
Mineral occurrences	All occurrences / MinDep-derived datasets, SARIG	Validation
Resources/reserves	South Australian Major Mines/Quarries and Mineral Projects	Independent resource/reserve validation
Drillholes/geochemistry	Mineral Drillholes, Petroleum and Geothermal Wells; Mineral drilling and geochemistry	Drillhole validation and geochemical classification

Supplementary Table S1. Data sources used in this study, including magnetic, geological, structural, mineral occurrence, resource/reserve, and drillhole datasets obtained primarily from SARIG and South Australian government data services.